

Patrick Carroll

Model and Simulation Engineer

☎ 843-437-6951 | ✉ patrickcarroll.it@outlook.com | 🔗 PCarroll9500 | 🐙 PCarroll9500.github.io

Security Clearance

Active Top-Secret SCI 2014 - Present

Summary

Model & Simulation Engineer specializing in AFSIM plugin development, simulation model integration, and CI/CD automation for defense applications. Combines 10+ years of technical and military experience across software development, engineering automation, systems administration, and RF signal analysis.

Skills

Programming Languages: C/C++, Python, MATLAB, HTML, CSS, JavaScript
Tools and Frameworks: TortoiseSVN, Git, Jira, Confluence, Jenkins, Groovy

Experience

Junior Model and Simulation Engineer	Radiance Technologies, Beaver Creek, Ohio	08/2024 - Present
- Develop and maintain CI/CD pipelines to build, test, and validate Simulink models into AFSIM plugins. - Package models and plugins to deploy across multiple government programs and agencies. - Partner with cross-functional engineering teams and government customers to integrate new models and features.		
Information Systems Technician	United States Navy Reserve, Cincinnati, Ohio	04/2019 - 04/2025
Supported the Navy part-time. Trained in Cisco Routers, Switches, End Devices, IPv4 and IPv6 addressing, network protocols, and troubleshooting. Qualified in Enlisted Information Warfare.		
System Software Support Engineer	MBS Professional, Mason, Ohio	01/2024 - 08/2024
- Developed Windows Preinstall scripts for Image Capture/Deployment. - Developed a VBA PowerPoint add-in that automated AI-driven text-to-speech video generation. - Developed VBA macros in MS Word for SQL database interactions.		
IT Systems Administration Assistant	Texas A&M Corpus Christi, Corpus Christi, Texas	08/2019 - 12/2021
Handled 700+ tickets dealing with the administration of User Accounts, E-mail, Listserv, Security Groups, Distribution Groups, Shared Mailboxes, File Systems, HPC and Active Directory.		
Submarine Electronics Technician	United States Navy, Norfolk, Virginia	04/2014 - 04/2019
- Analyzed and classified different types of RF signals. - Led teams of 2-3 in the maintenance and repair of critical systems. - Supported joint operations at sea.		

Projects

- Stock Bot**, Automated day-trading engine that scans for volatile stocks, analyzes news using GPT-4o, and executes trades via the Interactive Brokers API. Built with Python, and hosted on AWS.
- Document AI Organization & Scanner**, Document classification, renaming, meta-tagging, and organization using AI. Utilized Microsoft Azure AI Intelligence, OpenAI, Python, Java, and JavaScript.
- Battlezone**, Recreation of the 1980s tank simulator in C/C++ OpenGL, with iconic Green Vector 3D graphics, particle effects and collision detection.
- Comparing LLM Context Similarity**, Tested the similarities and accuracy of different OpenAI models with Jupyter Notebooks.

Education

B.S. in Computer Science (GPA 3.65) Texas A&M-Corpus Christi 04/2019 - 12/2023

Certifications

CompTIA Security+ 06/2022 - Present
Microsoft Certified: Azure AI Fundamentals 07/2023 - Present